

Norway

SLEIPNER

Volve F

F-11

15/9-F-11 B

Design: 15/9-F-11 B

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-11		
Well Position	+N/-S	4.65 m	Northing:
	+E/-W	-0.93 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.9566 N
			Longitude:
			1° 53' 14.8667 E
			Water Depth:
			91.00 m

Wellbore	15/9-F-11 B		
Magnetics	Model Name	Sample Date	Declination
			(°)
			Dip Angle
			(°)
			Field Strength
			(nT)
	MNETICREFERENCE	27.09.2012	-1.29
			71.67
			50,475

Design	15/9-F-11 B		
Audit Notes:			
Version:	1.0	Phase:	ACTUAL
		Tie On Depth:	2,570.00
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	4.65	-0.93
			96.40

Survey Program	Date	02.09.2016		
From	To	Survey (Wellbore)	Tool Name	Description
(m)	(m)			
150.00	2,570.00	F-11 T2 SDI Keeper Gyro (15/9-F-11 T2)	Keeper, cont	Gyro Tool from SDC
2,603.30	3,167.70	12 1/4" _BH_MWD (15/9-F-11 B)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
3,237.80	4,757.10	8 1/2" _BH_MWD_Multistation (15/9-F-11 B)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
4,770.00	4,770.00	Projection (15/9-F-11 B)	Blind Drilling	No survey recorded - blind drilling

Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
145.90	0.00	0.00	145.90	4.65	-0.93	0.00	0.000	0.000	0.000	
150.00	0.09	227.76	150.00	4.65	-0.93	0.00	0.659	0.659	0.000	
160.00	0.20	225.57	160.00	4.63	-0.95	-0.02	0.330	0.330	-6.570	
170.00	0.19	227.59	170.00	4.61	-0.97	-0.04	0.036	-0.030	6.060	
180.00	0.23	231.32	180.00	4.58	-1.00	-0.07	0.127	0.120	11.190	
190.00	0.23	233.02	190.00	4.56	-1.03	-0.09	0.020	0.000	5.100	
200.00	0.22	235.12	200.00	4.54	-1.06	-0.12	0.039	-0.030	6.300	
210.00	0.24	234.55	210.00	4.51	-1.10	-0.15	0.060	0.060	-1.710	

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Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
220.00	0.34	239.76	220.00	4.49	-1.14	-0.19	0.310	0.300	15.630
230.00	0.42	244.41	230.00	4.46	-1.20	-0.25	0.257	0.240	13.950
240.00	0.57	252.51	240.00	4.42	-1.28	-0.32	0.495	0.450	24.300
250.00	0.92	274.57	250.00	4.42	-1.41	-0.45	1.339	1.050	66.180
260.00	1.13	286.19	260.00	4.45	-1.58	-0.63	0.883	0.630	34.860
270.00	1.16	287.83	269.99	4.51	-1.77	-0.82	0.133	0.090	4.920
280.00	1.20	290.33	279.99	4.58	-1.97	-1.02	0.196	0.120	7.500
290.00	1.24	293.03	289.99	4.65	-2.16	-1.23	0.210	0.120	8.100
300.00	1.20	293.20	299.99	4.74	-2.36	-1.43	0.120	-0.120	0.510
310.00	1.19	294.42	309.99	4.82	-2.55	-1.63	0.082	-0.030	3.660
320.00	1.15	292.81	319.98	4.90	-2.74	-1.83	0.155	-0.120	-4.830
330.00	1.12	290.98	329.98	4.98	-2.92	-2.02	0.141	-0.090	-5.490
340.00	1.10	290.84	339.98	5.05	-3.10	-2.21	0.061	-0.060	-0.420
350.00	1.08	288.68	349.98	5.11	-3.28	-2.39	0.137	-0.060	-6.480
360.00	1.07	287.43	359.98	5.17	-3.46	-2.57	0.076	-0.030	-3.750
370.00	1.10	286.47	369.97	5.22	-3.64	-2.76	0.105	0.090	-2.880
380.00	1.19	289.32	379.97	5.29	-3.83	-2.96	0.319	0.270	8.550
390.00	1.33	294.68	389.97	5.37	-4.03	-3.17	0.549	0.420	16.080
400.00	1.73	303.22	399.97	5.50	-4.27	-3.41	1.378	1.200	25.620
410.00	1.86	305.68	409.96	5.68	-4.52	-3.69	0.453	0.390	7.380
420.00	1.89	306.23	419.96	5.87	-4.79	-3.97	0.105	0.090	1.650
430.00	1.96	306.14	429.95	6.07	-5.06	-4.27	0.210	0.210	-0.270
440.00	1.99	306.93	439.94	6.27	-5.34	-4.56	0.122	0.090	2.370
450.00	2.01	307.01	449.94	6.48	-5.62	-4.86	0.061	0.060	0.240
460.00	1.99	307.09	459.93	6.69	-5.89	-5.16	0.061	-0.060	0.240
470.00	2.03	307.66	469.93	6.91	-6.17	-5.46	0.134	0.120	1.710
480.00	2.06	308.33	479.92	7.13	-6.45	-5.77	0.115	0.090	2.010
490.00	2.05	307.74	489.91	7.35	-6.74	-6.07	0.070	-0.030	-1.770
500.00	2.10	310.18	499.91	7.57	-7.02	-6.38	0.305	0.150	7.320
510.00	2.11	309.44	509.90	7.81	-7.30	-6.69	0.087	0.030	-2.220
520.00	2.12	308.88	519.89	8.04	-7.59	-7.00	0.069	0.030	-1.680
530.00	2.09	308.61	529.89	8.27	-7.87	-7.31	0.095	-0.090	-0.810
540.00	2.13	310.13	539.88	8.51	-8.16	-7.62	0.206	0.120	4.560
550.00	2.13	309.28	549.87	8.74	-8.44	-7.93	0.095	0.000	-2.550
560.00	2.12	309.17	559.87	8.98	-8.73	-8.24	0.032	-0.030	-0.330
570.00	2.13	309.03	569.86	9.21	-9.02	-8.55	0.034	0.030	-0.420
580.00	2.16	309.02	579.85	9.45	-9.31	-8.86	0.090	0.090	-0.030
590.00	2.14	309.85	589.85	9.69	-9.60	-9.18	0.111	-0.060	2.490
600.00	2.18	309.02	599.84	9.93	-9.89	-9.50	0.152	0.120	-2.490
610.00	2.15	308.45	609.83	10.16	-10.18	-9.81	0.111	-0.090	-1.710
620.00	2.13	307.49	619.82	10.39	-10.48	-10.13	0.123	-0.060	-2.880
630.00	2.01	302.74	629.82	10.60	-10.77	-10.45	0.628	-0.360	-14.250
640.00	1.68	284.45	639.81	10.73	-11.06	-10.75	2.012	-0.990	-54.870

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Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
650.00	1.63	275.56	649.81	10.78	-11.35	-11.04	0.784	-0.150	-26.670
660.00	1.66	272.60	659.80	10.80	-11.63	-11.33	0.270	0.090	-8.880
670.00	1.79	276.80	669.80	10.83	-11.93	-11.63	0.544	0.390	12.600
680.00	2.11	265.05	679.79	10.83	-12.27	-11.96	1.532	0.960	-35.250
690.00	2.45	259.38	689.79	10.77	-12.66	-12.35	1.223	1.020	-17.010
700.00	2.77	256.59	699.78	10.68	-13.11	-12.78	1.033	0.960	-8.370
710.00	3.19	251.91	709.76	10.54	-13.61	-13.26	1.455	1.260	-14.040
720.00	4.01	244.94	719.74	10.30	-14.19	-13.81	2.784	2.460	-20.910
730.00	4.67	241.67	729.71	9.96	-14.87	-14.44	2.114	1.980	-9.810
740.00	5.38	239.18	739.67	9.53	-15.63	-15.15	2.228	2.130	-7.470
750.00	6.03	237.66	749.63	9.01	-16.47	-15.93	2.002	1.950	-4.560
760.00	6.89	236.29	759.56	8.39	-17.42	-16.80	2.621	2.580	-4.110
770.00	7.68	235.54	769.48	7.68	-18.47	-17.77	2.387	2.370	-2.250
780.00	8.37	235.75	779.38	6.89	-19.62	-18.83	2.072	2.070	0.630
790.00	9.33	235.86	789.26	6.03	-20.89	-19.99	2.880	2.880	0.330
800.00	10.21	236.82	799.12	5.09	-22.30	-21.29	2.685	2.640	2.880
810.00	10.79	237.99	808.95	4.11	-23.84	-22.71	1.854	1.740	3.510
820.00	11.33	239.39	818.77	3.11	-25.48	-24.23	1.809	1.620	4.200
830.00	11.93	242.49	828.56	2.13	-27.24	-25.87	2.598	1.800	9.300
840.00	12.31	246.06	838.34	1.22	-29.13	-27.65	2.521	1.140	10.710
850.00	12.52	249.24	848.10	0.41	-31.12	-29.53	2.145	0.630	9.540
860.00	12.79	252.71	857.86	-0.31	-33.19	-31.51	2.420	0.810	10.410
870.00	12.84	255.52	867.61	-0.91	-35.32	-33.56	1.876	0.150	8.430
880.00	12.75	256.87	877.36	-1.44	-37.47	-35.64	0.937	-0.270	4.050
890.00	12.55	257.29	887.12	-1.93	-39.61	-37.71	0.660	-0.600	1.260
900.00	12.18	258.31	896.89	-2.38	-41.70	-39.74	1.289	-1.110	3.060
910.00	11.40	260.19	906.68	-2.77	-43.71	-41.69	2.608	-2.340	5.640
920.00	10.91	261.87	916.49	-3.07	-45.62	-43.55	1.764	-1.470	5.040
930.00	10.76	262.94	926.31	-3.32	-47.48	-45.38	0.753	-0.450	3.210
940.00	10.58	264.05	936.14	-3.53	-49.32	-47.18	0.820	-0.540	3.330
950.00	10.53	264.50	945.97	-3.71	-51.14	-48.97	0.289	-0.150	1.350
960.00	10.52	265.36	955.80	-3.87	-52.96	-50.76	0.472	-0.030	2.580
970.00	10.48	266.01	965.63	-4.01	-54.78	-52.55	0.375	-0.120	1.950
980.00	10.36	267.69	975.47	-4.11	-56.59	-54.34	0.980	-0.360	5.040
990.00	10.31	270.33	985.31	-4.14	-58.38	-56.12	1.429	-0.150	7.920
1,000.00	10.29	272.52	995.14	-4.09	-60.17	-57.90	1.176	-0.060	6.570
1,010.00	10.34	273.32	1,004.98	-4.00	-61.95	-59.68	0.455	0.150	2.400
1,020.00	10.39	274.63	1,014.82	-3.88	-63.75	-61.48	0.723	0.150	3.930
1,030.00	10.48	275.34	1,024.65	-3.72	-65.55	-63.29	0.471	0.270	2.130
1,040.00	10.47	275.48	1,034.49	-3.55	-67.36	-65.11	0.082	-0.030	0.420
1,050.00	10.49	275.78	1,044.32	-3.37	-69.17	-66.93	0.174	0.060	0.900
1,060.00	10.66	277.07	1,054.15	-3.17	-71.00	-68.76	0.874	0.510	3.870
1,070.00	11.01	278.92	1,063.97	-2.90	-72.86	-70.64	1.480	1.050	5.550
1,080.00	11.46	281.15	1,073.78	-2.56	-74.78	-72.59	1.876	1.350	6.690

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Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,090.00	12.02	283.68	1,083.57	-2.13	-76.76	-74.61	2.282	1.680	7.590
1,100.00	13.04	287.32	1,093.33	-1.54	-78.85	-76.75	3.869	3.060	10.920
1,110.00	14.19	289.87	1,103.05	-0.79	-81.08	-79.05	3.891	3.450	7.650
1,120.00	15.14	291.09	1,112.73	0.10	-83.45	-81.50	2.997	2.850	3.660
1,130.00	16.29	291.76	1,122.35	1.09	-85.97	-84.12	3.493	3.450	2.010
1,140.00	18.12	291.94	1,131.91	2.19	-88.72	-86.97	5.492	5.490	0.540
1,150.00	19.37	291.62	1,141.37	3.38	-91.70	-90.07	3.763	3.750	-0.960
1,160.00	20.20	291.03	1,150.78	4.61	-94.86	-93.34	2.561	2.490	-1.770
1,170.00	20.94	290.28	1,160.15	5.85	-98.14	-96.75	2.357	2.220	-2.250
1,180.00	21.74	288.95	1,169.46	7.07	-101.57	-100.29	2.805	2.400	-3.990
1,190.00	22.06	288.24	1,178.74	8.26	-105.11	-103.93	1.246	0.960	-2.130
1,200.00	22.30	287.56	1,188.00	9.42	-108.70	-107.63	1.054	0.720	-2.040
1,210.00	22.53	286.61	1,197.24	10.54	-112.34	-111.38	1.287	0.690	-2.850
1,220.00	22.78	285.44	1,206.47	11.60	-116.05	-115.18	1.546	0.750	-3.510
1,230.00	22.91	284.77	1,215.69	12.61	-119.79	-119.01	0.872	0.390	-2.010
1,240.00	22.99	284.46	1,224.90	13.60	-123.57	-122.87	0.435	0.240	-0.930
1,250.00	23.10	284.53	1,234.10	14.58	-127.36	-126.75	0.340	0.330	0.210
1,260.00	23.02	284.46	1,243.30	15.56	-131.15	-130.63	0.254	-0.240	-0.210
1,270.00	22.40	284.23	1,252.52	16.51	-134.89	-134.45	1.879	-1.860	-0.690
1,280.00	22.06	284.22	1,261.78	17.44	-138.56	-138.20	1.020	-1.020	-0.030
1,290.00	21.49	284.19	1,271.07	18.35	-142.15	-141.87	1.710	-1.710	-0.090
1,300.00	20.95	283.85	1,280.39	19.23	-145.66	-145.46	1.662	-1.620	-1.020
1,310.00	20.83	283.60	1,289.73	20.08	-149.13	-149.00	0.448	-0.360	-0.750
1,320.00	20.85	283.68	1,299.08	20.92	-152.59	-152.53	0.104	0.060	0.240
1,330.00	21.02	283.73	1,308.42	21.76	-156.06	-156.07	0.513	0.510	0.150
1,340.00	21.31	283.42	1,317.74	22.61	-159.57	-159.65	0.933	0.870	-0.930
1,350.00	21.61	283.17	1,327.05	23.45	-163.13	-163.29	0.941	0.900	-0.750
1,360.00	21.97	282.58	1,336.33	24.28	-166.75	-166.97	1.264	1.080	-1.770
1,370.00	22.45	281.18	1,345.59	25.06	-170.44	-170.74	2.143	1.440	-4.200
1,380.00	23.22	278.29	1,354.81	25.71	-174.27	-174.61	4.081	2.310	-8.670
1,390.00	23.84	276.25	1,363.98	26.21	-178.23	-178.60	3.070	1.860	-6.120
1,400.00	24.53	274.57	1,373.10	26.60	-182.31	-182.70	2.924	2.070	-5.040
1,410.00	25.42	271.77	1,382.17	26.83	-186.52	-186.91	4.439	2.670	-8.400
1,420.00	26.38	269.11	1,391.16	26.86	-190.89	-191.25	4.521	2.880	-7.980
1,430.00	27.21	266.65	1,400.09	26.70	-195.39	-195.71	4.155	2.490	-7.380
1,440.00	27.94	264.45	1,408.95	26.34	-200.01	-200.26	3.759	2.190	-6.600
1,450.00	28.74	261.88	1,417.75	25.77	-204.72	-204.88	4.376	2.400	-7.710
1,460.00	29.48	260.21	1,426.49	25.01	-209.52	-209.57	3.297	2.220	-5.010
1,470.00	29.92	259.85	1,435.18	24.15	-214.40	-214.32	1.424	1.320	-1.080
1,480.00	30.40	259.60	1,443.82	23.26	-219.35	-219.13	1.488	1.440	-0.750
1,490.00	31.22	259.49	1,452.41	22.33	-224.38	-224.04	2.466	2.460	-0.330
1,500.00	31.87	259.45	1,460.93	21.37	-229.53	-229.04	1.951	1.950	-0.120
1,510.00	32.35	259.26	1,469.40	20.39	-234.75	-234.12	1.472	1.440	-0.570

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,520.00	32.72	259.31	1,477.84	19.39	-240.04	-239.26	1.113	1.110	0.150
1,530.00	33.21	259.32	1,486.23	18.38	-245.38	-244.46	1.470	1.470	0.030
1,540.00	33.41	259.40	1,494.58	17.37	-250.78	-249.71	0.614	0.600	0.240
1,550.00	33.61	259.49	1,502.92	16.36	-256.21	-255.00	0.618	0.600	0.270
1,560.00	33.95	259.53	1,511.23	15.34	-261.67	-260.32	1.022	1.020	0.120
1,570.00	34.05	259.37	1,519.52	14.32	-267.17	-265.66	0.403	0.300	-0.480
1,580.00	34.02	258.98	1,527.81	13.27	-272.67	-271.01	0.661	-0.090	-1.170
1,590.00	34.03	258.75	1,536.10	12.19	-278.16	-276.35	0.387	0.030	-0.690
1,600.00	33.97	258.48	1,544.39	11.08	-283.64	-281.67	0.487	-0.180	-0.810
1,610.00	34.00	258.45	1,552.68	9.97	-289.12	-286.99	0.103	0.090	-0.090
1,620.00	34.20	258.47	1,560.96	8.84	-294.61	-292.32	0.601	0.600	0.060
1,630.00	34.38	258.36	1,569.22	7.71	-300.13	-297.68	0.571	0.540	-0.330
1,640.00	34.44	258.17	1,577.47	6.56	-305.66	-303.05	0.369	0.180	-0.570
1,650.00	34.42	257.88	1,585.72	5.39	-311.19	-308.42	0.496	-0.060	-0.870
1,660.00	34.35	257.77	1,593.97	4.20	-316.71	-313.77	0.281	-0.210	-0.330
1,670.00	34.31	257.67	1,602.23	3.00	-322.23	-319.11	0.207	-0.120	-0.300
1,680.00	34.23	257.70	1,610.49	1.80	-327.73	-324.45	0.245	-0.240	0.090
1,690.00	34.23	257.66	1,618.76	0.60	-333.22	-329.78	0.068	0.000	-0.120
1,700.00	34.30	257.68	1,627.03	-0.60	-338.72	-335.11	0.213	0.210	0.060
1,710.00	34.41	257.59	1,635.28	-1.81	-344.23	-340.45	0.363	0.330	-0.270
1,720.00	34.04	257.50	1,643.55	-3.03	-349.73	-345.77	1.120	-1.110	-0.270
1,730.00	33.09	257.75	1,651.88	-4.21	-355.13	-351.01	2.880	-2.850	0.750
1,740.00	32.35	258.27	1,660.30	-5.33	-360.41	-356.14	2.375	-2.220	1.560
1,750.00	31.64	258.71	1,668.78	-6.39	-365.61	-361.18	2.242	-2.130	1.320
1,760.00	30.59	259.30	1,677.34	-7.38	-370.68	-366.11	3.280	-3.150	1.770
1,770.00	29.68	259.79	1,685.99	-8.29	-375.62	-370.91	2.828	-2.730	1.470
1,780.00	29.07	260.04	1,694.70	-9.15	-380.44	-375.62	1.867	-1.830	0.750
1,790.00	28.52	260.12	1,703.46	-9.98	-385.19	-380.24	1.654	-1.650	0.240
1,800.00	27.54	260.26	1,712.29	-10.78	-389.82	-384.75	2.947	-2.940	0.420
1,810.00	26.37	260.65	1,721.21	-11.53	-394.29	-389.11	3.550	-3.510	1.170
1,820.00	25.69	260.79	1,730.19	-12.24	-398.62	-393.34	2.048	-2.040	0.420
1,830.00	25.10	261.16	1,739.22	-12.91	-402.86	-397.47	1.833	-1.770	1.110
1,840.00	24.31	261.30	1,748.31	-13.55	-406.99	-401.50	2.376	-2.370	0.420
1,850.00	23.40	261.24	1,757.45	-14.16	-410.98	-405.41	2.731	-2.730	-0.180
1,860.00	22.79	260.74	1,766.65	-14.78	-414.86	-409.19	1.922	-1.830	-1.500
1,870.00	21.92	260.49	1,775.90	-15.40	-418.61	-412.85	2.626	-2.610	-0.750
1,880.00	20.98	260.58	1,785.21	-16.00	-422.22	-416.37	2.822	-2.820	0.270
1,890.00	20.23	261.02	1,794.57	-16.56	-425.69	-419.76	2.297	-2.250	1.320
1,900.00	19.65	261.39	1,803.97	-17.08	-429.06	-423.05	1.781	-1.740	1.110
1,910.00	19.02	261.79	1,813.41	-17.57	-432.34	-426.25	1.931	-1.890	1.200
1,920.00	18.16	262.11	1,822.88	-18.01	-435.49	-429.34	2.598	-2.580	0.960
1,930.00	17.32	262.45	1,832.41	-18.42	-438.51	-432.29	2.539	-2.520	1.020
1,940.00	16.56	262.81	1,841.97	-18.80	-441.40	-435.12	2.302	-2.280	1.080

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
1,950.00	15.62	263.53	1,851.58	-19.13	-444.15	-437.82	2.883	-2.820	2.160	
1,960.00	14.58	264.78	1,861.24	-19.39	-446.75	-440.36	3.269	-3.120	3.750	
1,970.00	13.75	265.49	1,870.93	-19.60	-449.18	-442.76	2.544	-2.490	2.130	
1,980.00	13.12	265.75	1,880.66	-19.78	-451.50	-445.05	1.899	-1.890	0.780	
1,990.00	12.11	266.00	1,890.42	-19.94	-453.68	-447.19	3.034	-3.030	0.750	
2,000.00	10.90	266.44	1,900.22	-20.07	-455.67	-449.16	3.640	-3.630	1.320	
2,010.00	10.04	267.49	1,910.05	-20.16	-457.48	-450.95	2.643	-2.580	3.150	
2,020.00	9.28	269.00	1,919.91	-20.22	-459.16	-452.61	2.403	-2.280	4.530	
2,030.00	8.48	271.02	1,929.79	-20.22	-460.70	-454.14	2.576	-2.400	6.060	
2,040.00	7.52	273.53	1,939.69	-20.16	-462.09	-455.53	3.064	-2.880	7.530	
2,050.00	6.87	275.05	1,949.61	-20.07	-463.34	-456.78	2.032	-1.950	4.560	
2,060.00	6.28	277.53	1,959.55	-19.95	-464.48	-457.93	1.964	-1.770	7.440	
2,070.00	5.66	282.49	1,969.49	-19.77	-465.50	-458.96	2.418	-1.860	14.880	
2,080.00	5.24	287.86	1,979.45	-19.52	-466.42	-459.90	1.981	-1.260	16.110	
2,090.00	4.77	293.02	1,989.41	-19.22	-467.24	-460.75	1.951	-1.410	15.480	
2,100.00	4.19	297.86	1,999.38	-18.89	-467.94	-461.49	2.076	-1.740	14.520	
2,110.00	3.37	306.50	2,009.36	-18.54	-468.50	-462.08	2.989	-2.460	25.920	
2,120.00	2.89	317.71	2,019.34	-18.18	-468.91	-462.52	2.327	-1.440	33.630	
2,130.00	2.79	328.96	2,029.33	-17.78	-469.20	-462.86	1.696	-0.300	33.750	
2,140.00	2.93	341.18	2,039.32	-17.33	-469.41	-463.12	1.873	0.420	36.660	
2,150.00	3.39	355.71	2,049.30	-16.80	-469.52	-463.28	2.760	1.380	43.590	
2,160.00	3.94	7.56	2,059.28	-16.16	-469.49	-463.33	2.800	1.650	35.550	
2,170.00	4.35	16.27	2,069.26	-15.46	-469.34	-463.26	2.250	1.230	26.130	
2,180.00	4.85	25.12	2,079.22	-14.71	-469.06	-463.06	2.600	1.500	26.550	
2,190.00	5.41	33.13	2,089.18	-13.93	-468.62	-462.71	2.724	1.680	24.030	
2,200.00	6.00	38.53	2,099.13	-13.13	-468.04	-462.22	2.391	1.770	16.200	
2,210.00	6.56	42.36	2,109.07	-12.30	-467.33	-461.61	2.097	1.680	11.490	
2,220.00	7.30	46.42	2,119.00	-11.44	-466.48	-460.86	2.661	2.220	12.180	
2,230.00	8.25	51.54	2,128.91	-10.55	-465.46	-459.95	3.524	2.850	15.360	
2,240.00	9.24	55.20	2,138.79	-9.65	-464.24	-458.83	3.406	2.970	10.980	
2,250.00	10.14	57.43	2,148.65	-8.72	-462.84	-457.54	2.925	2.700	6.690	
2,260.00	11.11	59.85	2,158.48	-7.76	-461.26	-456.09	3.202	2.910	7.260	
2,270.00	12.04	61.78	2,168.27	-6.78	-459.51	-454.45	3.022	2.790	5.790	
2,280.00	12.77	63.25	2,178.04	-5.79	-457.60	-452.67	2.386	2.190	4.410	
2,290.00	13.29	64.43	2,187.78	-4.80	-455.58	-450.77	1.752	1.560	3.540	
2,300.00	13.92	65.47	2,197.50	-3.80	-453.45	-448.76	2.027	1.890	3.120	
2,310.00	14.69	65.98	2,207.19	-2.79	-451.20	-446.64	2.341	2.310	1.530	
2,320.00	15.47	66.14	2,216.85	-1.73	-448.82	-444.39	2.343	2.340	0.480	
2,330.00	16.18	66.32	2,226.47	-0.63	-446.32	-442.03	2.135	2.130	0.540	
2,340.00	16.93	66.49	2,236.06	0.51	-443.71	-439.57	2.255	2.250	0.510	
2,350.00	17.72	66.96	2,245.60	1.69	-440.98	-436.98	2.407	2.370	1.410	
2,360.00	18.61	67.36	2,255.10	2.90	-438.10	-434.26	2.696	2.670	1.200	
2,370.00	19.52	67.72	2,264.55	4.14	-435.08	-431.40	2.753	2.730	1.080	
2,380.00	20.43	68.12	2,273.95	5.43	-431.92	-428.39	2.761	2.730	1.200	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,390.00	21.25	68.32	2,283.30	6.75	-428.61	-425.26	2.469	2.460	0.600
2,400.00	21.92	68.35	2,292.60	8.10	-425.20	-422.01	2.010	2.010	0.090
2,410.00	22.61	68.54	2,301.85	9.50	-421.67	-418.67	2.081	2.070	0.570
2,420.00	23.43	68.67	2,311.05	10.92	-418.03	-415.21	2.465	2.460	0.390
2,430.00	24.36	68.72	2,320.20	12.39	-414.26	-411.62	2.791	2.790	0.150
2,440.00	25.31	68.55	2,329.27	13.92	-410.35	-407.90	2.858	2.850	-0.510
2,450.00	26.29	68.14	2,338.28	15.53	-406.30	-404.06	2.988	2.940	-1.230
2,460.00	27.23	67.77	2,347.20	17.22	-402.13	-400.10	2.864	2.820	-1.110
2,470.00	28.23	67.35	2,356.06	19.00	-397.83	-396.03	3.057	3.000	-1.260
2,480.00	29.08	67.17	2,364.83	20.85	-393.40	-391.84	2.563	2.550	-0.540
2,490.00	30.08	67.26	2,373.53	22.76	-388.85	-387.53	3.003	3.000	0.270
2,500.00	31.02	67.18	2,382.14	24.73	-384.17	-383.09	2.823	2.820	-0.240
2,510.00	31.64	67.35	2,390.68	26.74	-379.37	-378.55	1.879	1.860	0.510
2,520.00	32.15	67.45	2,399.17	28.77	-374.49	-373.93	1.538	1.530	0.300
2,530.00	32.87	67.45	2,407.60	30.83	-369.53	-369.23	2.160	2.160	0.000
2,540.00	34.06	67.96	2,415.95	32.92	-364.43	-364.39	3.668	3.570	1.530
2,550.00	35.36	68.34	2,424.17	35.04	-359.14	-359.37	3.954	3.900	1.140
2,560.00	36.24	68.58	2,432.28	37.19	-353.70	-354.20	2.673	2.640	0.720
2,570.00	36.93	68.78	2,440.31	39.36	-348.15	-348.93	2.101	2.070	0.600
2,603.30	38.69	72.50	2,466.62	46.11	-328.89	-330.54	2.595	1.586	3.351
2,643.40	41.55	75.42	2,497.28	53.23	-304.06	-306.66	2.561	2.140	2.185
2,683.90	46.01	77.51	2,526.52	59.76	-276.82	-280.32	3.473	3.304	1.548
2,724.40	50.83	78.14	2,553.39	66.14	-247.22	-251.61	3.587	3.570	0.467
2,764.70	54.97	77.33	2,577.69	72.98	-215.82	-221.17	3.119	3.082	-0.603
2,805.20	59.75	75.82	2,599.53	80.90	-182.66	-189.10	3.664	3.541	-1.119
2,844.70	63.05	73.92	2,618.44	89.96	-149.19	-156.85	2.808	2.506	-1.443
2,885.60	62.94	73.91	2,637.01	100.06	-114.18	-123.18	0.081	-0.081	-0.007
2,925.80	61.95	79.05	2,655.62	108.39	-79.54	-89.69	3.480	-0.739	3.836
2,966.30	61.79	84.11	2,674.72	113.62	-44.23	-55.17	3.307	-0.119	3.748
3,005.30	61.93	86.52	2,693.12	116.43	-9.96	-21.43	1.638	0.108	1.854
3,045.90	61.86	90.52	2,712.25	117.36	25.83	14.03	2.608	-0.052	2.956
3,086.20	62.08	93.76	2,731.19	116.03	61.37	49.50	2.135	0.164	2.412
3,126.90	62.08	97.11	2,750.25	112.62	97.17	85.45	2.182	0.000	2.469
3,167.70	62.72	101.07	2,769.16	106.91	132.86	121.56	2.623	0.471	2.912
3,237.80	61.97	108.75	2,801.73	90.96	192.80	182.90	2.928	-0.321	3.287
3,278.30	65.40	113.86	2,819.70	77.76	226.59	217.96	4.238	2.541	3.785
3,318.20	69.26	116.66	2,835.07	62.04	259.87	252.78	3.492	2.902	2.105
3,359.20	70.23	117.13	2,849.27	44.64	294.18	288.81	0.780	0.710	0.344
3,399.60	71.72	119.84	2,862.44	26.42	327.74	324.19	2.201	1.106	2.012
3,439.70	71.30	124.16	2,875.16	6.28	359.98	358.48	3.081	-0.314	3.232
3,480.00	71.64	128.17	2,887.97	-16.27	390.82	391.64	2.842	0.253	2.985
3,520.50	74.08	126.55	2,899.91	-39.74	421.58	424.83	2.140	1.807	-1.200
3,561.00	77.83	125.23	2,909.74	-62.77	453.41	459.02	2.935	2.778	-0.978

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,601.40	81.42	123.47	2,917.01	-85.19	486.21	494.12	2.960	2.666	-1.307	
3,641.30	85.04	119.76	2,921.72	-105.94	519.95	529.95	3.883	2.722	-2.789	
3,681.70	88.67	118.68	2,923.93	-125.63	555.15	567.13	2.812	2.696	-0.802	
3,721.90	90.01	117.94	2,924.39	-144.69	590.53	604.42	1.142	1.000	-0.552	
3,762.10	89.09	117.33	2,924.71	-163.34	626.15	641.89	0.824	-0.687	-0.455	
3,802.40	89.10	116.96	2,925.35	-181.72	662.00	679.57	0.276	0.007	-0.275	
3,842.70	88.48	117.88	2,926.20	-200.28	697.77	717.18	0.826	-0.462	0.685	
3,884.00	83.95	112.08	2,928.93	-217.67	735.09	756.21	5.338	-3.291	-4.213	
3,924.50	82.59	108.99	2,933.67	-231.78	772.75	795.21	2.486	-1.007	-2.289	
3,964.40	80.05	105.02	2,939.70	-243.32	810.46	833.96	3.515	-1.910	-2.985	
4,004.80	77.79	101.56	2,947.46	-252.44	849.03	873.31	3.029	-1.678	-2.569	
4,044.90	75.06	98.96	2,956.88	-259.38	887.38	912.20	2.783	-2.042	-1.945	
4,085.00	76.09	97.76	2,966.87	-265.03	925.80	951.01	1.162	0.771	-0.898	
4,124.90	77.05	95.80	2,976.13	-269.61	964.34	989.81	1.605	0.722	-1.474	
4,165.40	79.04	93.43	2,984.52	-272.79	1,003.83	1,029.41	2.263	1.474	-1.756	
4,205.70	80.10	91.39	2,991.82	-274.46	1,043.42	1,068.95	1.689	0.789	-1.519	
4,245.90	80.09	88.85	2,998.74	-274.54	1,083.02	1,108.31	1.867	-0.007	-1.896	
4,286.40	78.98	85.96	3,006.09	-272.74	1,122.80	1,147.64	2.260	-0.822	-2.141	
4,327.30	77.61	81.29	3,014.39	-268.30	1,162.59	1,186.69	3.501	-1.005	-3.425	
4,367.60	75.14	78.75	3,023.89	-261.52	1,201.16	1,224.26	2.599	-1.839	-1.891	
4,408.30	70.74	77.64	3,035.83	-253.56	1,239.24	1,261.21	3.336	-3.243	-0.818	
4,448.30	66.71	76.16	3,050.34	-245.12	1,275.53	1,296.34	3.194	-3.022	-1.110	
4,488.30	61.74	75.23	3,067.73	-236.23	1,310.43	1,330.03	3.780	-3.727	-0.697	
4,528.60	57.15	73.30	3,088.21	-226.84	1,343.82	1,362.17	3.634	-3.417	-1.437	
4,569.20	52.85	71.54	3,111.49	-216.81	1,375.52	1,392.55	3.351	-3.177	-1.300	
4,609.60	49.15	69.40	3,136.91	-206.33	1,405.11	1,420.79	3.012	-2.748	-1.589	
4,649.90	44.55	67.61	3,164.47	-195.57	1,432.46	1,446.77	3.559	-3.424	-1.333	
4,690.30	39.91	65.44	3,194.38	-184.78	1,457.37	1,470.32	3.611	-3.446	-1.611	
4,730.30	37.45	64.41	3,225.60	-174.20	1,480.01	1,491.64	1.907	-1.845	-0.772	
4,757.10	36.80	64.18	3,246.97	-167.18	1,494.58	1,505.34	0.744	-0.728	-0.257	
4,770.00	36.80	64.18	3,257.30	-163.81	1,501.54	1,511.88	0.000	0.000	0.000	

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
202.00	202.00	30"	30.000	36.000	
1,357.70	1,334.20	20"	20.000	26.000	
2,570.68	2,440.85	14"	14.000	17.500	
3,192.40	2,780.54	9 5/8"x10 3/4"	9.625	12.250	
4,768.00	3,255.70	7"	7.000	8.500	

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54.90m (MI)
Site:	Volve F	MD Reference:	MI @ 54.90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Design Annotations					
Measured Depth (m)	Vertical Depth (m)	Local Coordinates		Comment	
		+N/-S (m)	+E/-W (m)		
4,770.00	3,257.30	-163.81	1,501.54	Projection to TD	

Checked By: _____	Approved By: _____	Date: _____
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